Verify Trig Identities (VTI) Practice
AP Precalculus Unit 4 Wkst C **Name**_____

Verify the identities in "Required:" column or sides (29 required problems). At the completion of each section evaluate your level of competency. If you need additional practice choose problems from the "more practice" (problems are on the last page). If you are staring at the problem with no clue where to start. 1) TRY SOMETHING. If you still need help 2) watch the indicated video on Canvas. Label problems 1.a, 1.b, 1.c, 2.a, 2.b, etc

	Required:	What's the verdict?	Your choice
Section 1: Quotient/Reciprocal	**a.** $\csc\theta\cos\theta=\cot\theta$	I've got this. No additional practice needed. I'm ok but could use more practice I need help! (additional instruction)	More practice 3, 4,7,8,11,12 Additional instruction Section 1 Video on Canvas
	b. $\frac{\sec\theta}{\csc\theta}=\tan\theta$		
	c. $\cot\theta\sec\theta=\csc\theta$		
Section 2: Pythagorean	**a.** $\sin^3\theta+\cos^2\theta\sin\theta=\sin\theta$	I've got this. No additional practice needed. I'm ok but could use more practice I need help! (additional instruction)	More practice 1,2,5,6,9,10,15 Additional instruction Section 2 Video on Canvas
	b. $\tan^2\theta+6=\sec^2\theta+5$		
	c. $\frac{\sin\theta}{\csc\theta}+\frac{\cos\theta}{\sec\theta}=1$		
Section 3:Distributive Property	**a.** $\cos\theta(1+\sec\theta)=\cos\theta+1$	I've got this. No additional practice needed. I'm ok but could use more practice I need help! (additional instruction)	More practice 13,14,16,17 Additional instruction Section 3 Video on Canvas
	b. $\csc\theta-\cos\theta\cot\theta=\sin\theta$		
	c. $(\csc\theta+\cot\theta)(\csc\theta-\cot\theta)=1$		
Section 4: Rationalize Numerator/Denomi	**a.** $\frac{\csc\theta+\cot\theta}{\sin\theta+\tan\theta}=\cos\theta\csc^2\theta$	I've got this. No additional practice needed. I'm ok but could use more practice I need help! (additional instruction)	More practice 18, 19, 21, 22 Additional instruction Section 4 Video on Canvas
	b. $\frac{\sec\theta+\tan\theta}{\cos\theta+\cot\theta}=\tan\theta\sec\theta$		
	c. $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}}=\frac{1-\sin\theta}{\cos\theta}$		
Section 5: Common Denominator	**a.** $\frac{\sec\theta-\cos\theta}{\sin\theta}+\frac{\csc\theta-\sin\theta}{\cos\theta}=\tan\theta+\cot\theta$	I've got this. No additional practice needed. I'm ok but could use more practice I need help! (additional instruction)	More practice 20 Additional instruction Section 5 Video on Canvas
	b. $\frac{1}{\sec\theta-1}+\frac{1}{\sec\theta+1}=2\cot\theta\csc\theta$		
	c. $\frac{1}{1-\sin\theta}+\frac{1}{1+\sin\theta}=2\sec^2\theta$		

Section 6: Combination of skills: Level 1
Required: Pick a side. If you feel you need additional practice choose problems from the other side.
Label problems as 6.A.a, 6.A.b, etc OR 6.B.a, 6.B.b, etc

Side A	Side B
a. $\dfrac{\sec\theta}{\csc^2\theta} = \sec\theta - \cos\theta$	**a.** $\dfrac{1-2\csc\theta}{\cot\theta} = \tan\theta - 2\sec\theta$
b. $\sin\theta + \cos\theta\cot\theta = \csc\theta$	**b.** $\cos\theta\left(\csc\theta - \sec\theta\right) = \cot\theta - 1$
c. $\dfrac{\sin\theta+\cos\theta}{\sin\theta\cos\theta} = \sec\theta + \csc\theta$	**c.** $\dfrac{\left(1+\sin\theta\right)^2}{\cos^2\theta} = \dfrac{1+\sin\theta}{1-\sin\theta}$
d. $\csc\theta\cos^2\theta + \sin\theta = \csc\theta$	**d.** $\tan^2\theta - \tan^2\theta\sin^2\theta = \sin^2\theta$
e. $\sin\theta\left(\dfrac{\cot\theta}{\sec\theta} + \csc\theta\right) = \cos^2\theta + 1$	**e.** $\csc\theta - \sin\theta = \cot\theta\cos\theta$
f. $\dfrac{\csc^2\theta}{\csc^2\theta - 1} = \sec^2\theta$	**f.** $\dfrac{2\cos^2\theta - \sin^2\theta + 1}{\cos\theta} = 3\cos\theta$
g. $\sin^4\theta - \cos^4\theta = 1 - 2\cos^2\theta$	**g.** $\left(\sin\theta - \cos\theta\right)^2 + 2\sin\theta - \cos\theta = \left(1+2\sin\theta\right)\left(1-\cos\theta\right)$

Section 7: Combination of skills: Level 2
Required: Pick a side. If you feel you need additional practice choose problems from the other side.
Label problems as 7.A.h, 7.A.i, etc OR 7.B.h, 7.B.i, etc

Side A	Side B
h. $\sin(x+y) + \sin(x-y) = 2\sin x\cos y$	**h.** $\cos(x+y) + \cos(x-y) = 2\cos x\cos y$
i. $\cos(x+y)\cos(x-y) = \cos^2 x - \sin^2 y$	**i.** $\sin(x+y)\sin(x-y) = \sin^2 x - \sin^2 y$
j. $\cos^2 2x - \sin^2 2x = \cos 4x$	**j.** $\cos^4 x - \sin^4 x = \cos 2x$
k. $\dfrac{\cos 3x}{\cos x} = 1 - 4\sin^2 x$	**k.** $\pm\sqrt{\dfrac{2\tan x}{\tan x+\sin x}} = \sec\dfrac{x}{2}$
l. $\cos 3x = \cos^3 x - 3\sin^2 x\cos x$	**l.** $\sin 4x = 4\sin x\cos x(1 - 2\sin^2 x)$
m. $\dfrac{\sin x\cos y+\cos x\sin y}{\cos x\cos y-\sin x\sin y} = \dfrac{\tan x+\tan y}{1-\tan x\tan y}$	**m.** $\dfrac{\tan x+\tan y}{1-\tan x\tan y} = \dfrac{\cot x+\cot y}{\cot x\cot y-1}$
n. $\tan^5 x = \tan^3 x\sec^2 x - \tan^3 x$	**n.** $\sin^4 x + \cos^4 x = 1 - 2\cos^2 x + 2\cos^4 x$

More Practice

1. $\cos^3\theta+\sin^2\theta\cos\theta=\cos\theta$

2. $\csc^2\theta-\cos^2\theta\csc^2\theta=1$

3. $\sec\theta\sin\theta=\tan\theta$

4. $\dfrac{\csc\theta}{\sec\theta}=\cot\theta$

5. $\dfrac{\sec^2\theta-1}{\tan\theta}=\tan\theta$

6. $\dfrac{\cot\theta}{\csc^2\theta-1}=\tan\theta$

7. $\sec\theta\sin\theta\cot\theta=1$

8. $\cot\theta\csc\theta\tan^2\theta=\sec\theta$

9. $\cos^2\theta-\sin^2\theta=2\cos^2\theta-1$

10. $\cos^2\theta-\sin^2\theta=1-2\sin^2\theta$

11. $\cot\theta\sin\theta=\cos\theta$

12. $\dfrac{\tan\theta}{\sec\theta}=\sin\theta$

13. $\sin\theta\left(1+\csc\theta\right)=\sin\theta+1$

14. $\left(1+\tan\theta\right)^2=\sec^2\theta+2\tan\theta$

15. $\left(1+\tan^2\theta\right)\cos^2\theta=1$

16. $\cos\theta=\sec\theta-\sin\theta\tan\theta$

17. $\left(\sec\theta+\tan\theta\right)\left(\sec\theta-\tan\theta\right)=1$

18. $\dfrac{\sec\theta+\tan\theta}{\cos\theta+\cot\theta}=\sin\theta\sec^2\theta$

19. $\dfrac{1+\sec\theta}{\tan\theta+\sin\theta}=\csc\theta$

20. $\dfrac{1}{1-\cos\theta}+\dfrac{1}{1+\cos\theta}=2\csc^2\theta$

21. $\sqrt{\dfrac{\sec\theta+\tan\theta}{\sec\theta-\tan\theta}}=\dfrac{1+\sin\theta}{\cos\theta}$

22. $\sqrt{\dfrac{1-\cos\theta}{1+\cos\theta}}=\dfrac{1-\cos\theta}{\sin\theta}$